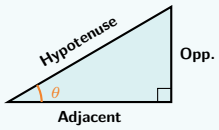
TRIGONOMETRY

Secondary School Revision Cheatsheet • Ratios · Identities · Graphs · Rules

PrepNAce
Learn smart. Prep sharp.

1 Right-Angled Triangles & SOH-CAH-TOA

$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}} \qquad \cos\theta = \frac{\text{Adj}}{\text{Hyp}} \qquad \tan\theta = \frac{\text{Opp}}{\text{Adj}}$$



Pythagoras: $a^2 + b^2 = c^2$ (c = hypotenuse). **Find an angle:** $\theta = \sin^{-1}\left(\frac{\text{Opp}}{\text{Hyp}}\right)$ (same for $\cos^{-1}, \tan^{-1}$).

Reciprocals: $\csc\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}$

Example: Opp $= 5$, Hyp $= 13$ $\Rightarrow$ $\sin\theta = \frac{5}{13}$ $\Rightarrow$ $\theta = \sin^{-1}\left(\frac{5}{13}\right) \approx 22.6°$

2 Exact Values (Special Angles)

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef.

Memory trick: sin of $0, 30, 45, 60, 90$ is $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cos is the same list backwards.

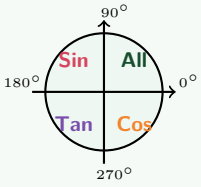
3 Key Identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta \qquad 1 + \cot^2\theta = \csc^2\theta$$

Rearrange: $\sin^2\theta = 1 - \cos^2\theta$, $\cos^2\theta = 1 - \sin^2\theta$. Complementary: $\sin(90° - \theta) = \cos\theta$.

4 Any Angle: CAST Rule & Related Angles



Angle	sin	cos	tan
$180° - \theta$	$+\sin\theta$	$-\cos\theta$	$-\tan\theta$
$180° + \theta$	$-\sin\theta$	$-\cos\theta$	$+\tan\theta$
$360° - \theta$	$-\sin\theta$	$+\cos\theta$	$-\tan\theta$
$90° - \theta$	$\cos\theta$	$\sin\theta$	$\frac{1}{\tan\theta}$

CAST = which ratio is *positive* in each quadrant (anticlockwise from Q4: **C, A, S, T**).

5 Radians, Arcs & Sectors

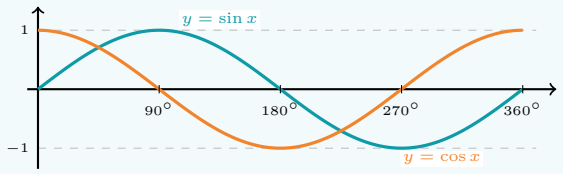
Convert: π rad $= 180°$; rad $=$ deg $\times \frac{\pi}{180}$; deg $=$ rad $\times \frac{180}{\pi}$.

Deg	$30°$	$45°$	$60°$	$90°$	$180°$	$360°$
Rad	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π

$$l = r\theta \qquad A = \tfrac{1}{2}r^2\theta \quad (\theta \text{ in radians})$$

Exam traps: set your calculator to **DEG** (or RAD when asked); $\sin^{-1}$ gives only *one* answer, so use CAST for the rest; label Opp/Adj *relative to* θ; $\tan 90°$ is undefined.

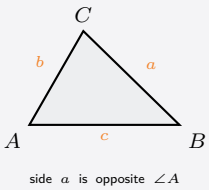
6 Graphs of Trig Functions



	$\sin x$	$\cos x$	$\tan x$
Period	$360°$	$360°$	$180°$
Range	$[-1, 1]$	$[-1, 1]$	all reals
Special	passes $(0, 0)$	starts at 1	asymptotes $x = 90°, 270°$

Transformations: $y = a\sin(bx) + c$ $\Rightarrow$ amplitude $|a|$, period $\frac{360°}{b}$, midline $y = c$.

7 Any Triangle: Sine & Cosine Rules



side a is opposite $\angle A$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Use for: 2 angles + 1 side, or 2 sides + angle opposite one (*ambiguous case*: may give 2 triangles).

$$a^2 = b^2 + c^2 - 2bc\cos A \qquad \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Use for: 2 sides + included angle (SAS), or 3 sides (SSS).

Area of a triangle $= \frac{1}{2}ab\sin C$ (2 sides + angle between).

Example: $b = 7$, $c = 8$, $A = 60°$ $\Rightarrow$ $a^2 = 49 + 64 - 2(7)(8)\cos 60° = 57$ $\Rightarrow$ $a \approx 7.55$

8 Compound & Double-Angle Formulae

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$$
$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$
$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

$$\sin 2A = 2\sin A\cos A \qquad \tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

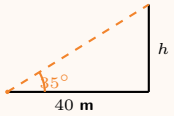
9 Solving Trig Equations ($0° \le \theta \le 360°$)

1. Isolate the ratio, e.g. $\sin\theta = k$.
2. Find the acute **basic angle** $\alpha = \sin^{-1}|k|$ (ignore sign).
3. Use **CAST** for the quadrants; then solve:

Equation	Solutions
$\sin\theta = +k$	$\alpha,\ 180° - \alpha$
$\cos\theta = +k$	$\alpha,\ 360° - \alpha$
$\tan\theta = +k$	$\alpha,\ 180° + \alpha$

Example: $2\sin\theta = 1$ $\Rightarrow$ $\sin\theta = \frac{1}{2}$, $\alpha = 30°$ $\Rightarrow$ $\theta = 30°,\ 150°$.

10 Real-Life Applications



Elevation: angle *up* from horizontal.
Depression: angle *down* from horizontal.
Bearings: clockwise from North, 3 digits (e.g. $045°$).

Example: from 40 m away the top of a tower is at $35°$ elevation. $h = 40\tan 35° \approx 28.0$ m.